TED(15)4013 Reg No…………

REVISION-2015 Signature…………

FORTH SEMESTER DIPLOMA EXAMINATION IN CIVIL ENGINEERING

QUANTITY SURVEYING-I (Time-3 hours)

(Maximum marks-100)

PART-A

(Maximum marks-10) Mark

I. Answer all questions in one or two sentences. Each questions carries 2 marks.

1.State any two duties of a quantity surveyor.

2. List any four types of approximate estimate.

3. Define contingencies.

4. State trapezoidal rule.

5. State tolerance of wastage of materials. (5X2=10)

PART-B

(Maximum marks-30)

II. Answer any five of the following questions. Each carries 6 marks.

1. Explain methods of taking out quantities.
2. A tank is excavated in a level ground to a depth of 12m with the uniform sides slope. The top of tank at ground level is in rectangular shape of size 40x20m while the bottom Is 24x14m. Compute the volume of earth work by prismoidal formula.
3. Compute the quantity of brick work of compound wall of inside dimension 23x20m Wall height 1.45m and 4.5m gate opening
4. Determine the under side plastering of roof slab from figure 1.
5. Find the quantity of earth work of a masonry well of diameter 2m and a depth of 12m .The steining of wall is of 30cm thick first class brick work.
6. Explain cost of materials at source and laboure cost.
7. List any four general rules of measurements of completed works. (5X6=30)

PART-C

(Answer one full question from each module. Each question carries 15 marks)

MODULE-1

III. a. Define an estimate and explain necessary data for preparation of an estimate. 7

b. The areas with contours at the site of a proposed reservoir are given below.

Find reservoir capacity using trapezoidal , simpson's rule and compare the results. 8

Contour in m	101	102	103	104	105	106	107
Area in m^2	528	910	1500	1750	2100	2800	3100

OR

IV. a. Distinguish between supplementary estimate and revised estimate. 7

b. The details of road embankment are as follows, formation width is12m side slope 2:1

RL of formation is 53m. Compute the quantity of earth work using prismoidal rule. 8

Chainage in m	0	30	60	90	120	150	180
Ground level	49.1	47.6	49.9	48.5	49	49.6	49.4

MODULE-2

V. Determine the quantities of following items from figure 1.

a. Earth work excavation in ordinary soil . 7

b. Random rubble masonary in foundation and basement. 8

OR

VI. Compute the quantities of items from figure 1.

a. Brick work in CM 1:6 for super structure. 7

b. Plain cement concrete 1:4:8. 8

MODULE-3

VII. a. Determine the quantity of flooring with polished granite tiles of 40mm over a bed Of 1:4:8 cement concrete and tiles set in CM from the figure 1. 7

b. Determine the quantity of RCC of roof slab, sunshade, lintel from given figure 1. 8

OR

VIII. a. Compute the quantity of plastering both inside and outside from figure1. 7

b. Determine the quantity of painting two coats with approved quality enamel paint over a priming coat to new wood work. 8

MODULE-4

IX. a. Explain standard data book and schedule of rates . 7

b. Workout the rate per unit of random rubble masonry in CM 1:4 for the basement and foundation from the following data. Add profit to contractor.

Materials

$1m^3$ Rubble @ R_s2500/m^3

$0.3m^3$ Dry sand @ R_s2000/m^3

108kg cement @ R_s3800/ton

Labour

0.7 Rubble mason @ R_s850/each /day

.35 man @ R_s700/each/day

.7 women @ R_S650/each/day 8

OR

X. a. Explain the procedure for preparing rate analysis 7

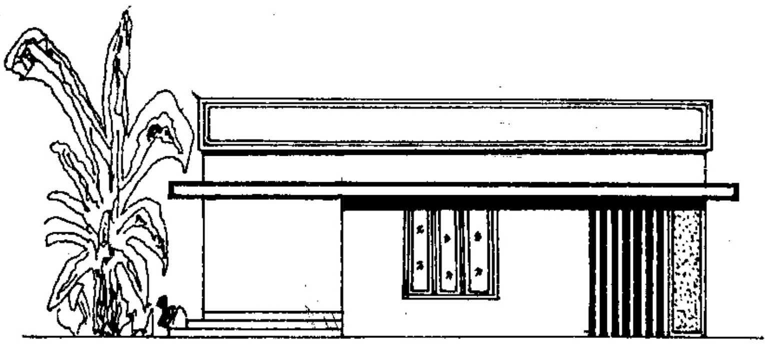
b. Prepare a specimen abstract estimate with the following items . The quantities and rate given below.

1. Earth work excavation	$15.8m^3$	@R_s400/m^3
2. RR masonry	$7.1m^3$	@R_s1800/m^3
3. Brick work CM 1:6	$16.8m^3$	@R_s1500/m^3
4. RCC work for roofing	$8.6m^3$	@R_s2500/m^3
5. Plastering	$186m^2$	@R_s220/m^2
6. flooring	$52m^2$	@R_s300/m^2

7.Electrication charge 8% of cost

8. Water supply and sanitary 8% of cost

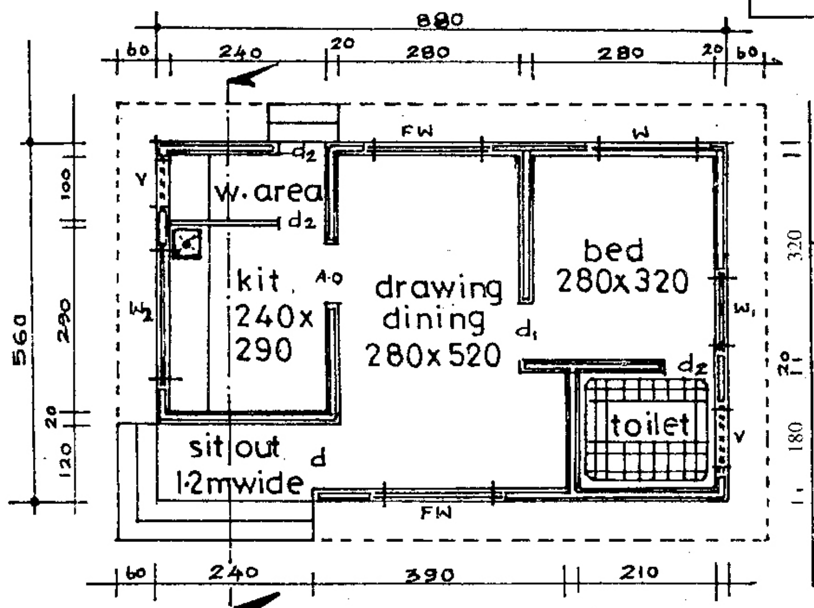
9. Unforeseen items 2% of cost. 8



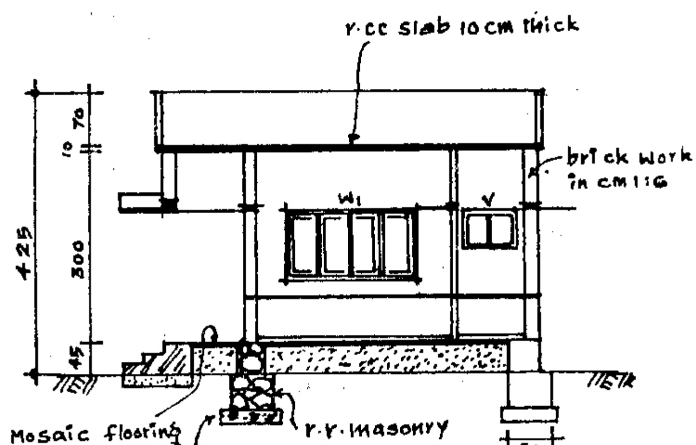
ELEVATION

Specification

Door	d	100	x	200
"	d_1	90	x	200
"	d_2	75	x	200
Window	w	150	x	140
"	w_1	105	x	140
"	w_2	180	x	10[illegible]
"	fw	165	x	18[illegible]
Ventilation	V	80	x	60



PLAN



SECTION